

Why Attribution Reporting Feels Wrong

A Guided Sandbox for Identity Stitching, Journey Reconstruction, and Attribution Modelling

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Introduction

At some point, most marketers experience the same uncomfortable feeling:

The numbers technically work. **But the story does not fully make sense.**

Different platforms report different outcomes.

Revenue appears duplicated.

“Direct” traffic keeps growing mysteriously.

Meta looks weak under last-click reporting but clearly influences demand.

Search appears dominant, yet often captures intent created elsewhere.

And despite all the dashboards, attribution models, and reporting layers, it becomes difficult to confidently answer a simple question:

What actually influenced this conversion?

This project started from that tension.

Not from a desire to build a production attribution platform.

Not from a desire to prove a “correct” attribution model exists.

But from a desire to better understand:

- how attribution systems work underneath the surface,
- how identity stitching shapes visibility,
- why attribution models tell different stories,
- and where interpretation begins to diverge from observable truth.

The deeper this exploration went, the clearer something became:

Attribution is not really a reporting problem.

It is an interpretation problem.

The same conversion data can produce completely different strategic narratives depending on:

- (SQL) stitching quality,
- journey construction,
- attribution philosophy,
- and measurement assumptions.

This sandbox was built to explore those ideas transparently using:

- lightweight tooling,
- synthetic datasets,
- deterministic modelling,
- and step-by-step validation.

The goal is not certainty. **The goal is developing better reasoning around ambiguity.**

The Problem This Sandbox Is Trying To Solve

Imagine this situation:

- Meta says it generated the conversion.
- Google Ads also claims the conversion.
- GA4 shows “Direct”.
- CRM revenue does not match platform reporting.
- Finance asks why ROAS dropped.
- Marketing says awareness campaigns are working.
- Search campaigns appear to drive most purchases.
- Nobody fully trusts the numbers.

This is extremely common in modern marketing.

The problem is not usually:

- one platform being “wrong”.

The problem is that:

- customer journeys are fragmented,
- platforms only see part of the journey,
- attribution models use different assumptions,
- and revenue often gets duplicated across reporting systems.

This sandbox exists to help marketers understand:

- how those problems happen,
- how attribution systems are constructed,
- and why different attribution models produce different strategic narratives from the same underlying conversions.

Rather than treating attribution as a black box, this workflow breaks the system down step-by-step using simplified datasets and lightweight tooling.

Who This Sandbox Is For

This project is for marketers who:

- feel their reporting does not fully explain customer behaviour,
- suspect platform attribution is misleading,
- struggle reconciling frontend metrics with backend outcomes,
- rely heavily on spreadsheets or platform dashboards,
- want to understand how attribution systems actually work,
- or are beginning to move toward analytics engineering and marketing intelligence workflows.

No prior attribution modelling experience is required.

The workflow is intentionally designed to:

- start simple,
- remain commercially grounded,
- and progressively introduce more advanced analytical thinking.

What You Will Build

Throughout this project, you will progressively construct the foundational layers behind a modern marketing measurement system.

Layer	Purpose
Identity Layer	Stitch users across systems
Validation Layer	Reconcile backend truth
Journey Layer	Reconstruct ordered customer touchpoints
Attribution Layer	Apply deterministic attribution models
Interpretation Layer	Analyse behavioural implications
Insight Layer	Compare strategic narratives across models

This is fundamentally different from simple dashboard reporting.

The workflow demonstrates how:

- identity quality,
- measurement assumptions,
- and attribution philosophy

shape marketing interpretation itself.

Estimated Completion Time

This sandbox can be approached at multiple depth levels depending on how deeply you want to explore the concepts.

Learning Depth	Estimated Time
Basic walkthrough	~2–3 hours
Full guided exploration	~5–8 hours
Deep experimentation & extension	10+ hours

Most of the technical workflow is lightweight and fast to run locally.

The majority of the learning comes from:

- interpreting outputs,
- validating assumptions,
- understanding attribution ambiguity,
- and thinking critically about how measurement systems shape marketing interpretation.

This project is intentionally designed to be:

- explored gradually,
- revisited iteratively,
- and extended over time.

Important Limitation

This sandbox does NOT prove marketing causality.

Attribution models are:

- interpretive frameworks,
- not objective truth systems.

They distribute conversion credit using assumptions about behavioural influence.

That distinction matters.

A strong analyst understands:

- how the models work,
 - what assumptions they encode,
 - what blind spots they contain,
 - and how those assumptions change strategic decision-making.
-

Quickstart

Before you get started, make sure everything is set up correctly and you've installed all dependencies.

Read:

None

[setup_guide.pdf](#)

Once you've gone through this, go straight into Step 1 - Stitching Data.

The recommended workflow is:

1. Run the script
 2. Observe the output
 3. Compare against walkthrough (this document)
-

Step 1 — Stitch Identity Data

Before we can analyse marketing performance, we first need to connect the data together.

Right now, the customer journey is fragmented across multiple systems:

- website events,
- ad platforms,
- and backend sales data.

This step attempts to connect those pieces together so we can follow a user from:

- marketing interaction,
all the way through to:
- conversion outcome.

Think of this as rebuilding the customer journey from scattered puzzle pieces.

Script

Python

`01_stitch_data.py`

Goal

Connect frontend marketing events with backend outcomes using shared identifiers.

What This Script Is Doing

This script combines:

- frontend behavioural events,
- ad platform activity,
- and backend conversion outcomes

into a unified stitched dataset.

The stitching logic attempts to connect users across systems using:

- `order_id` (strongest identifier; actual purchases / transactions),
- `Quote_id`, (leads)
- and `stitched_user_key`.

Each stitched row is assigned a `match_quality` label to indicate how confidently the event was connected to a backend outcome.

Output

```
match_quality  count_star()
0  order_id_match      15
1  quote_id_match     136
2  no_backend_match   1491
3  user_id_match      547
```

What You're Seeing

The output shows the number of stitched rows by match quality.

Example:

- `user_id_match = 547`
- `quote_id_match = 136`
- `order_id_match = 15`
- `no_backend_match = 1491`

This tells us:

- most events were connected through weaker identity stitching methods,
- while only a small number connected directly to a confirmed order.

Why This Matters

Attribution quality depends heavily on identity quality.

If users cannot be stitched reliably:

- journeys fragment,
- conversions become disconnected from marketing touchpoints,
- and attribution becomes distorted.

This is one of the biggest hidden problems in marketing analytics.

Important Limitation

A stitched relationship does NOT prove causality.

It only means:

- the systems observed a relationship between identifiers,
- not that the marketing activity definitively caused the conversion.

Strategic Interpretation

Weak identity stitching creates attribution ambiguity.

A marketer may:

- over-credit some channels,
- under-credit others,
- or lose visibility entirely

depending on how well users are connected across systems.

Step 2 — Validate Data Structure

Now that the data has been connected together, we need to check whether everything looks correct.

This step is about making sure:

- events are appearing properly,
- timestamps make sense,
- and the dataset behaves the way we expect.

Before doing attribution analysis, we first need confidence that the underlying data is usable.

Otherwise:

- incorrect data
will produce:
- misleading conclusions.

Script

Python

`02_validate_data.py`

Goal

Inspect schema, validate joins, and understand event quality.

What This Script Is Doing

This script validates the stitched dataset by:

- inspecting table structure,
- reviewing sample rows,
- checking event quality,
- and analysing how events relate to match quality.

It helps confirm the stitched dataset behaves as expected before attribution modelling begins.

Output

```
=== TABLE STRUCTURE ===
column_name column_type null key default extra
0 stitched_user_key VARCHAR YES None None None
1 event_timestamp TIMESTAMP YES None None None
2 source VARCHAR YES None None None
3 medium VARCHAR YES None None None
4 event_name VARCHAR YES None None None
5 gclid VARCHAR YES None None None
6 fbclid VARCHAR YES None None None
7 quote_id VARCHAR YES None None None
8 order_id VARCHAR YES None None None
9 backend_status VARCHAR YES None None None
10 revenue DOUBLE YES None None None
11 match_quality VARCHAR YES None None None

=== SAMPLE ROWS ===
stitched_user_key event_timestamp source medium ... order_id backend_status revenue match_quality
0 ui_6a793a7ae7 2026-03-28 01:49:58 Google Ads cpc ... None nurture 0.00 user_id_match
1 ui_6a793a7ae7 2026-03-28 01:50:18 Google Ads cpc ... None nurture 0.00 user_id_match
2 ui_6a793a7ae7 2026-03-28 01:50:38 Google Ads cpc ... None nurture 0.00 user_id_match
3 1730074215.1084793 2026-03-28 02:44:58 Google Ads cpc ... None nurture 0.00 quote_id_match
4 1730074215.1084793 2026-03-28 02:45:28 Google Ads cpc ... None nurture 0.00 quote_id_match
5 1323408297.1809764 2026-03-17 22:02:24 Google Ads cpc ... None lost 0.00 quote_id_match
6 1323408297.1809764 2026-03-17 22:02:54 Google Ads cpc ... None lost 0.00 quote_id_match
7 ui_6a52e1d11b 2026-03-15 07:27:45 Meta paid_social ... None lost 0.00 quote_id_match
8 ui_6a52e1d11b 2026-03-15 07:28:15 Meta paid_social ... None lost 0.00 quote_id_match
9 ui_c86239d075 2026-03-06 07:31:32 Meta paid_social ... None won 108.29 user_id_match

[10 rows x 12 columns]

=== MATCH QUALITY BY EVENT ===
event_name match_quality row_count
0 session_start no_backend_match 589
1 page_view no_backend_match 589
2 page_view user_id_match 139
3 session_start user_id_match 139
4 scroll no_backend_match 109
5 view_item no_backend_match 100
6 select_content no_backend_match 94
7 generate_lead user_id_match 78
8 form_submit user_id_match 78
9 form_submit quote_id_match 69
10 generate_lead quote_id_match 65
11 purchase user_id_match 42
12 select_content user_id_match 30
13 scroll user_id_match 22
14 view_item user_id_match 19
15 purchase order_id_match 15
16 generate_lead no_backend_match 7
17 form_submit no_backend_match 3
18 purchase quote_id_match 2
```

What You're Seeing

The output displays:

- the stitched schema,
- sample event rows,
- and event counts grouped by match quality.

Examples include:

- `session_start`
- `page_view`
- `generate_lead`
- `purchase`

matched across:

- `user_id_match`
- `quote_id_match`
- `order_id_match`
- `and no_backend_match.`

Why This Matters

Bad joins create misleading attribution outputs.

If:

- event timestamps are incorrect,
- identifiers are broken,
- or joins multiply rows unexpectedly,

then attribution models become unreliable.

Validation is mandatory before interpretation.

Important Limitation

A clean schema does not guarantee correct attribution.

It only confirms:

- the data structure appears logically consistent.

Behavioural interpretation still requires caution.

Strategic Interpretation

Many attribution problems are actually:

- data engineering problems,
- stitching problems,
- or schema design problems.

Marketers often trust attribution outputs without validating the underlying joins first.

Step 3 — Validate Join Quality

Now we need to check something extremely important:

Are we accidentally duplicating revenue?

This is one of the most common problems in marketing reporting.

A single conversion can appear multiple times across different touchpoints, making revenue look much larger than it really is.

This step compares:

- the true backend revenue,
against:
- the stitched marketing dataset,

to make sure the numbers still reconcile correctly.

In simple terms:

we are checking whether the attribution system is accidentally “making up” extra revenue through duplication.

Script

Python

`03_check_backend_truth_vs_stitched.py`

Goal

Detect:

- duplicate revenue,
- broken joins,
- inflated conversions,
- and mismatched identifiers.

What This Script Is Doing

This script compares:

- backend source-of-truth revenue
vs
- stitched touchpoint visibility.

It validates whether attribution-ready datasets reconcile correctly back to backend outcomes.

The script also checks:

- duplicated revenue across touchpoints,
- repeated conversion rows,
- and revenue inflation risk.

Output

```
=== RAW TABLE ROW COUNTS ===
      table_name  rows
0      ga4_events 2189
1 backend_outcomes 150
2  stitched_events 2189

=== BACKEND SOURCE OF TRUTH ===
status backend_rows distinct_order_ids distinct_quote_ids distinct_backend_user_ids backend_revenue
0      won           75                75                75                75          28353.55
1      lost           29                 0                29                29           0.00
2  nurture           25                 0                25                25           0.00
3      open           21                 0                21                21           0.00

=== BACKEND WON CONVERSIONS ONLY ===
won_rows won_order_ids won_revenue
0         75             75      28353.55

=== STITCHED EVENTS VIEW OF WON CONVERSIONS ===
stitched_won_event_rows stitched_won_order_ids stitched_won_quote_ids stitched_won_users repeated_stitched_revenue
0                    406                 51                 75                 92          147128.59

=== MATCH QUALITY SUMMARY ===
match_quality rows users order_ids repeated_revenue
0 no_backend_match 1491  462         0              NaN
1  user_id_match   547  104         36      119277.20
2  quote_id_match  136   72         0      21494.15
3  order_id_match   15   15         15      6357.24

=== ORDER-ID DUPLICATION CHECK ===
If this returns an empty table, there are no duplicated won orders at the order_id match level.
Revenue can still be repeated across user_id or quote_id stitched touchpoints, so use backend_outcomes as source of truth.
Empty DataFrame
Columns: [order_id, backend_status, true_order_revenue, matched_touchpoint_rows, inflated_revenue_if_summed]
Index: []
```

What You're Seeing

Several critical relationships appear here.

Backend Source Of Truth

The backend system reports:

- 75 won conversions
- \$28,353.55 total revenue

This becomes the financial source-of-truth for the entire project.

Stitched Event View

The stitched dataset shows:

- 406 stitched touchpoint rows
- 92 stitched users
- \$147,128.59 repeated stitched revenue

This is NOT real revenue growth.

The same revenue is being repeated across multiple observable touchpoints.

Match Quality Summary

The output also shows:

- user_id_match repeated revenue = \$119,277
- quote_id_match repeated revenue = \$21,494
- order_id_match repeated revenue = \$6,357

This demonstrates how attribution systems repeatedly distribute the same conversion value across journeys.

Why This Matters

Revenue reconciliation is mandatory before attribution interpretation.

Without reconciliation:

- attribution models can massively inflate performance,
- duplicate revenue silently,
- and mislead strategic decision-making.

This is one of the most common analytical mistakes in marketing reporting.

A dashboard may appear correct while silently multiplying revenue across touchpoints underneath the surface.

Important Limitation

This script does NOT prove:

- a channel caused a conversion.

It only validates:

- whether stitched touchpoints reconcile logically against backend outcomes.

Strategic Interpretation

The backend system should remain:

- the financial source-of-truth.

Attribution systems should be treated as:

- interpretive frameworks,
- not accounting systems.

Step 4 — Build Conversion Journeys

Now that the data has been validated, we can begin reconstructing customer journeys.

Instead of looking at:

- isolated clicks,
or:
- single events,

we now start looking at:

- the sequence of interactions users had before converting.

This helps answer questions like:

- How many touchpoints happened before purchase?
- Which channels appeared earlier?
- Which channels appeared later?
- What paths do customers commonly take?

We are essentially rebuilding the observable journey step-by-step.

Script

Python

`04_backend_first_conversion_journeys.py`

Goal

Reconstruct chronological user journeys.

What This Script Is Doing

This script builds ordered customer journeys using:

- backend-confirmed conversions,
- stitched touchpoints,
- and event timestamps.

Each conversion path is reconstructed chronologically so touchpoint sequencing can be analysed.

Output

```
C:\Users\kyler\k2\Projects\local-attribution-sandbox>python scripts/04c_backend_first_conversion_journeys.py

=== BACKEND WON CONVERSIONS ===
backend_won_rows  distinct_conversion_keys  backend_revenue
0                75                        28353.55

=== CONVERSION JOURNEYS CREATED ===
conversion_journeys  deduped_revenue  avg_touchpoints
0                   75             28353.55             5.413333

=== SAMPLE JOURNEYS ===
conversion_key  backend_user_id  quote_id  order_id  conversion_revenue  touchpoint_count  journey_path
0 order_T0S87MVZ  be_e348310acf  quote_VB9EKT1  order_T0S87MVZ  617.17  2 Meta / paid_social / generate_lead > Meta / pa...
1 order_EICUUSYX  ui_3bc06b36eb  quote_VBKDYJPJZ  order_EICUUSYX  614.06  6 Meta / paid_social / session_start > Meta / pa...
2 order_ZJ396YFI  be_6c7e434a54  quote_3EJFB0VB  order_ZJ396YFI  683.77  3 Meta / paid_social / generate_lead > Meta / pa...
3 order_1U25ZCFP  ui_e76d4332a1  quote_FLA8Q0M6  order_1U25ZCFP  600.98  5 Direct / (none) / session_start > Direct / (no...
4 order_9P4MM8DD  be_3f0u21cf85  quote_109JMB5X  order_9P4MM8DD  599.19  3 Meta / paid_social / generate_lead > Meta / pa...
5 order_W9JPKPF3  ui_e538014dde  quote_7H318N07  order_W9JPKPF3  597.84  5 Direct / (none) / session_start > Direct / (no...
6 order_FL140FW  be_1c3bbd969c  quote_D82795E5  order_FL140FW  593.39  2 Direct / (none) / generate_lead > Direct / (no...
7 order_V724HK1Q  ui_202d612166  quote_9TU8PV2K  order_V724HK1Q  593.02  5 Direct / (none) / session_start > Direct / (no...
8 order_UU4YTN02  ui_e7cbc5356d  quote_7L8V1Z0C  order_UU4YTN02  588.51  10 Meta / paid_social / session_start > Meta / pa...
9 order_D7UAYBDT  ui_68c1da6ff9  quote_NPMZKXCB  order_D7UAYBDT  582.71  5 Meta / paid_social / session_start > Meta / pa...

=== JOURNEYS WITHOUT TOUCHPOINTS ===
conversions_without_touchpoints
0
```

What You're Seeing

The output displays:

- conversion keys,
- journey paths,
- touchpoint counts,
- and revenue per conversion.

The dataset produced:

- 75 valid conversion journeys
- \$28,353.55 reconciled revenue
- 5.4 average touchpoints per conversion

Example journey:

Meta / paid_social / generate_lead → Google Ads / cpc / purchase

Why This Matters

Attribution is fundamentally journey analysis.

Without journey reconstruction:

- attribution becomes isolated event reporting,
- rather than behavioural sequence analysis.

The average touchpoint count (~5.4) suggests:

- users rarely convert immediately,
- and multiple interactions influence behaviour before conversion.

Important Limitation

Observed journeys are only:

- observable tracked touchpoints.

Offline behaviour,
cross-device activity,
dark social,
and untracked exposures remain invisible.

Strategic Interpretation

This step reframes marketing performance away from:

- isolated clicks

towards:

- cumulative behavioural influence across time.

Step 5 — First-Touch & Last-Touch Attribution

Now we can begin testing attribution models.

The first models are intentionally simple:

- first-touch attribution,
- and last-touch attribution.

First-touch gives all the credit to:

- the first interaction.

Last-touch gives all the credit to:

- the final interaction before conversion.

Even though these models are simple, they often tell completely different stories about which channels appear most valuable.

This is where attribution starts becoming interesting.

Script

Python

```
05_build_touchpoint_attribution.py
```

Goal

Compare:

- discovery channels,
- vs
- conversion capture channels.

What This Script Is Doing

This script generates:

- first-touch attribution
- and last-touch attribution

using the reconstructed conversion journeys.

First-touch:

- gives full credit to the first observable interaction.

Last-touch:

- gives full credit to the final observable interaction before conversion.

Output

```
C:\Users\kyler\k2\Projects\local-attribution-sandbox>python scripts/05_build_touchpoint_attribution.py

=== TOUCHPOINT ATTRIBUTION CREATED ===
touchpoint_rows  conversions
0               406         75

=== FIRST TOUCH REVENUE BY CHANNEL ===
source      medium  first_touch_revenue
0      Meta  paid_social  13664.88
1 Google Ads  cpc        7107.66
2      Direct  (none)     6296.44
3      Unknown (not set)  907.61
4      Organic  organic    376.96

=== LAST TOUCH REVENUE BY CHANNEL ===
source      medium  last_touch_revenue
0 Google Ads  cpc        10917.85
1      Direct  (none)     7456.31
2      Meta  paid_social  6587.40
3      Organic  organic    2000.14
4      Unknown (not set)  1391.85

=== SAMPLE TOUCHPOINTS ===
conversion_key  source      medium  event_name  revenue  first_touch_revenue  last_touch_revenue
0 order_2XCV96Q3 Google Ads  cpc        purchase  455.81      0.00      455.81
1 order_2XCV96Q3 Google Ads  cpc        form_submit 455.81      0.00      0.00
2 order_2XCV96Q3 Google Ads  cpc        generate_lead 455.81      455.81     0.00
3 order_AP8UBK6S Direct      (none)     form_submit 327.01      0.00      327.01
4 order_AP8UBK6S Direct      (none)     generate_lead 327.01      0.00      0.00
5 order_AP8UBK6S Direct      (none)     scroll 327.01      0.00      0.00
6 order_AP8UBK6S Direct      (none)     page_view 327.01      0.00      0.00
7 order_AP8UBK6S Direct      (none)     session_start 327.01      327.01     0.00
8 order_IP3M7ZUM Google Ads  cpc        purchase  239.41      0.00      239.41
9 order_IP3M7ZUM Google Ads  cpc        form_submit 239.41      0.00      0.00
10 order_IP3M7ZUM Google Ads  cpc        generate_lead 239.41      0.00      0.00
11 order_IP3M7ZUM Google Ads  cpc        scroll 239.41      0.00      0.00
12 order_IP3M7ZUM Google Ads  cpc        page_view 239.41      0.00      0.00
13 order_IP3M7ZUM Google Ads  cpc        session_start 239.41      239.41     0.00
14 order_R801KJWV Organic      organic     form_submit 445.97      0.00      445.97
15 order_R801KJWV Organic      organic     generate_lead 445.97      0.00      0.00
16 order_R801KJWV Meta  paid_social  scroll 445.97      0.00      0.00
17 order_R801KJWV Meta  paid_social  page_view 445.97      0.00      0.00
18 order_R801KJWV Meta  paid_social  session_start 445.97      445.97     0.00
19 order_X0100VHM Google Ads  cpc        form_submit 134.19      0.00      134.19
20 order_X0100VHM Google Ads  cpc        generate_lead 134.19      0.00      0.00
21 order_X0100VHM Meta  paid_social  select_content 134.19      0.00      0.00
22 order_X0100VHM Meta  paid_social  page_view 134.19      0.00      0.00
23 order_X0100VHM Meta  paid_social  session_start 134.19      134.19     0.00
24 order_ZRMPGX44 Google Ads  cpc        purchase  443.02      0.00      443.02
```

What You're Seeing

The same conversions produce completely different channel valuations.

Examples:

Meta / paid_social

- First-touch: ~\$13.6k
- Last-touch: ~\$6.6k

Google Ads / cpc

- First-touch: ~\$7.1k
- Last-touch: ~\$10.9k

Why This Matters

Different attribution philosophies create different strategic narratives.

Meta appears significantly stronger under first-touch attribution, suggesting:

- discovery,
- awareness,
- and demand generation behaviour.

Google Ads appears stronger under last-touch attribution, suggesting:

- conversion capture,
- intent resolution,
- and lower-funnel behaviour.

Important Limitation

Neither model proves causality.

Both models:

- oversimplify behavioural complexity,
- and force full conversion credit onto a single touchpoint.

Strategic Interpretation

Last-click reporting can undervalue:

- upper-funnel activity,
- awareness channels,
- and demand generation.

Meanwhile:

- first-touch attribution can undervalue lower-funnel conversion capture activity.

The “best” model depends on the behavioural question being asked.

Step 6 — Linear Attribution

So far, each model has given 100% of the credit to a single touchpoint.

But real customer behaviour is usually more complicated than that.

People often interact with:

- multiple ads,
- multiple channels,
- and multiple sessions before converting.

Linear attribution tries to reflect that by sharing conversion credit evenly across the entire journey.

Instead of:

- “one winner gets all the credit,”

the model assumes:

- every touchpoint contributed something.

Script

Python

`06_linear_attribution.py`

Goal

Split revenue evenly across all touchpoints.

What This Script Is Doing

This model distributes conversion revenue equally across every touchpoint within a journey.

Example:

- 5 touchpoints
→ each receives 20% attribution credit.

Output

```
C:\Users\kyler\k2\Projects\local-attribution-sandbox>python scripts/06_linear_attribution.py

=== LINEAR ATTRIBUTION CREATED ===
touchpoint_rows conversions
0          406          75

=== LINEAR ATTRIBUTION BY CHANNEL ===
source medium linear_revenue
0 Google Ads cpc 9843.84
1 Meta paid_social 8963.64
2 Direct (none) 7102.49
3 Organic organic 1227.76
4 Unknown (not set) 1215.72

=== SAMPLE LINEAR ATTRIBUTION ===
conversion_key source medium revenue touchpoint_count linear_attribution_revenue
0 order_SIFPWIOX Meta paid_social 108.29 5 21.66
1 order_SIFPWIOX Meta paid_social 108.29 5 21.66
2 order_SIFPWIOX Meta paid_social 108.29 5 21.66
3 order_SIFPWIOX Meta paid_social 108.29 5 21.66
4 order_SIFPWIOX Meta paid_social 108.29 5 21.66
5 order_UU4YTN02 Meta paid_social 588.51 10 58.85
6 order_UU4YTN02 Meta paid_social 588.51 10 58.85
7 order_UU4YTN02 Meta paid_social 588.51 10 58.85
8 order_UU4YTN02 Google Ads cpc 588.51 10 58.85
9 order_UU4YTN02 Google Ads cpc 588.51 10 58.85
10 order_UU4YTN02 Google Ads cpc 588.51 10 58.85
11 order_UU4YTN02 Google Ads cpc 588.51 10 58.85
12 order_UU4YTN02 Google Ads cpc 588.51 10 58.85
13 order_UU4YTN02 Google Ads cpc 588.51 10 58.85
14 order_UU4YTN02 Google Ads cpc 588.51 10 58.85
15 order_FLI40FWW Direct (none) 593.39 2 296.70
16 order_FLI40FWW Direct (none) 593.39 2 296.70
17 order_P0RNP6J Meta paid_social 558.29 8 69.79
18 order_P0RNP6J Meta paid_social 558.29 8 69.79
19 order_P0RNP6J Meta paid_social 558.29 8 69.79
20 order_P0RNP6J Direct (none) 558.29 8 69.79
21 order_P0RNP6J Direct (none) 558.29 8 69.79
22 order_P0RNP6J Direct (none) 558.29 8 69.79
23 order_P0RNP6J Direct (none) 558.29 8 69.79
24 order_P0RNP6J Direct (none) 558.29 8 69.79

=== VALIDATION CHECK ===
total_linear_revenue
0 28353.45
```

What You're Seeing

The model produced:

- perfectly reconciled revenue,
- while distributing attribution across all observable journey interactions.

Example:

- 10-touchpoint journeys distribute revenue evenly across all 10 interactions.

Why This Matters

Linear attribution assumes:

- journeys are collaborative,
- not purely winner-take-all.

This often increases visibility into:

- mid-funnel influence,
- repeated exposure,
- and assistive channel behaviour.

Important Limitation

Linear attribution assumes every touchpoint contributes equally.

In reality:

- some touchpoints may exert minimal behavioural influence,
- while others may be highly decisive.

Strategic Interpretation

Linear attribution is often useful when marketers suspect customer behaviour is cumulative rather than immediate.

Instead of rewarding only:

- the first interaction,
or:
- the final conversion touchpoint,

this model highlights channels that consistently participate throughout the customer journey.

In this dataset, channels like:

- Meta,
- Direct,
- and Google Ads

all maintained meaningful participation across journeys, suggesting users repeatedly interacted with multiple touchpoints before converting.

This is especially common in:

- higher consideration purchases,
- longer buying cycles,
- and multi-session decision-making behaviour.

For marketers, this model can help surface:

- assistive channel behaviour,
 - repeated exposure patterns,
 - and channels quietly supporting conversion pathways underneath the final click.
-

Step 7 — Position-Based Attribution

Linear attribution treats every touchpoint equally.

But marketers often believe:

- the first interaction,
and:
- the final interaction

matter more than the middle steps.

Position-based attribution reflects that idea.

This model gives:

- extra credit to the beginning of the journey,
- extra credit to the end of the journey,
- and less credit to the middle interactions.

The goal is to balance:

- discovery,
with:
- conversion completion

Script

Python

`07_position_based_attribution.py`

Goal

Weight:

- first-touch heavily,
- last-touch heavily,
- middle touches lightly.

What This Script Is Doing

This model applies:

- heavier weighting to journey initiation,
- heavier weighting to conversion completion,
- and lighter weighting to middle interactions.

This attempts to balance:

- demand generation,
- and conversion capture.

Output

```
C:\Users\kyler\k2\Projects\local-attribution-sandbox>python scripts/07_position_based_attribution.py

=== POSITION-BASED ATTRIBUTION CREATED ===
touchpoint_rows conversions
0          406          75

=== POSITION-BASED REVENUE BY CHANNEL ===
source medium position_based_revenue
0      Meta paid_social 9823.89
1 Google Ads cpc 9226.34
2 Direct (none) 6934.59
3 Organic organic 1201.63
4 Unknown (not set) 1167.11

=== SAMPLE POSITION-BASED ATTRIBUTION ===
conversion_key source medium revenue touchpoint_position total_touchpoints position_based_revenue
0 order_1493685E Meta paid_social 162.96 1 11 65.18
1 order_1493685E Meta paid_social 162.96 2 11 3.62
2 order_1493685E Google Ads cpc 162.96 3 11 3.62
3 order_1493685E Google Ads cpc 162.96 4 11 3.62
4 order_1493685E Google Ads cpc 162.96 5 11 3.62
5 order_1493685E Google Ads cpc 162.96 6 11 3.62
6 order_1493685E Google Ads cpc 162.96 7 11 3.62
7 order_1493685E Google Ads cpc 162.96 8 11 3.62
8 order_1493685E Google Ads cpc 162.96 9 11 3.62
9 order_1493685E Google Ads cpc 162.96 10 11 3.62
10 order_1493685E Google Ads cpc 162.96 11 11 65.18
11 order_AJJ050KH Google Ads cpc 469.02 1 3 187.61
12 order_AJJ050KH Google Ads cpc 469.02 2 3 93.80
13 order_AJJ050KH Google Ads cpc 469.02 3 3 187.61
14 order_CWEQ4F1S Direct (none) 276.93 1 6 110.77
15 order_CWEQ4F1S Direct (none) 276.93 2 6 13.85
16 order_CWEQ4F1S Direct (none) 276.93 3 6 13.85
17 order_CWEQ4F1S Direct (none) 276.93 4 6 13.85
18 order_CWEQ4F1S Direct (none) 276.93 5 6 13.85
19 order_CWEQ4F1S Direct (none) 276.93 6 6 110.77
20 order_D7UAYBDT Meta paid_social 582.71 1 5 233.08
21 order_D7UAYBDT Meta paid_social 582.71 2 5 38.85
22 order_D7UAYBDT Meta paid_social 582.71 3 5 38.85
23 order_D7UAYBDT Meta paid_social 582.71 4 5 38.85
24 order_D7UAYBDT Meta paid_social 582.71 5 5 233.08

=== VALIDATION CHECK ===
total_position_based_revenue
0 28353.55
```

What You're Seeing

Revenue weighting shifts toward:

- the beginning,
- and end

of the customer journey.

Middle touchpoints receive reduced attribution value.

Why This Matters

Position-based attribution encodes a behavioural assumption:

That:

- journey initiation matters,
- journey completion matters,
- but middle interactions are less influential.

This creates a compromise between:

- first-touch,
- and last-touch philosophies.

Important Limitation

The weighting logic is still arbitrary.

The model assumes:

- first and last interactions matter most

without behavioural proof.

Strategic Interpretation

Position-based attribution reflects a common marketing belief:

That:

- discovery matters,
and:
- conversion completion matters,

more than the interactions in between.

In practice, this model often creates a more balanced view between:

- awareness activity,
- and lower-funnel conversion capture.

Compared to linear attribution, middle journey interactions lose influence, while channels appearing earlier or later in the journey gain additional value.

For marketers, this model is often psychologically appealing because it rewards:

- demand generation,
without:
- completely ignoring conversion-driving activity.

However, the weighting structure is still subjective.

The model assumes the beginning and end of the journey deserve more importance — even though real customer behaviour may not always follow that pattern.

Step 8 — Time-Decay Attribution

Time-decay attribution introduces a different idea:

The closer an interaction is to conversion, the more important it may have been.

This model gradually increases attribution credit as touchpoints move closer to the final purchase.

In practice, this often increases the perceived value of:

- branded search,
- direct traffic,
- and lower-funnel campaigns.

At the same time, earlier awareness interactions may receive less credit.

Script

Python

```
08_time_decay_attribution.py
```

Goal

Weight recent touchpoints more heavily.

What This Script Is Doing

This model gradually increases attribution credit as touchpoints move closer to conversion.

Earlier touchpoints receive:

- less weight.

Later touchpoints receive:

- more weight.

Output

```
C:\Users\kyler\k2\Projects\local-attribution-sandbox>python scripts/08_time_decay_attribution.py

=== TIME DECAY ATTRIBUTION CREATED ===
touchpoint_rows conversions
0 406 75

=== TIME DECAY REVENUE BY CHANNEL ===
source medium time_decay_revenue
0 Google Ads cpc 10521.37
1 Meta paid_social 7643.10
2 Direct (none) 7321.16
3 Organic organic 1552.43
4 Unknown (not set) 1315.49

=== SAMPLE TIME DECAY ATTRIBUTION ===
conversion_key source medium revenue touchpoint_position total_touchpoints decay_weight time_decay_revenue
0 order_85S0I99N Google Ads cpc 169.44 1 5 0.0667 11.30
1 order_85S0I99N Google Ads cpc 169.44 2 5 0.1333 22.59
2 order_85S0I99N Google Ads cpc 169.44 3 5 0.2000 33.89
3 order_85S0I99N Organic organic 169.44 4 5 0.2667 45.18
4 order_85S0I99N Organic organic 169.44 5 5 0.3333 56.48
5 order_K9WAM0KJ Meta paid_social 228.80 1 7 0.0357 8.17
6 order_K9WAM0KJ Meta paid_social 228.80 2 7 0.0714 16.34
7 order_K9WAM0KJ Google Ads cpc 228.80 3 7 0.1071 24.51
8 order_K9WAM0KJ Google Ads cpc 228.80 4 7 0.1429 32.69
9 order_K9WAM0KJ Google Ads cpc 228.80 5 7 0.1786 40.86
10 order_K9WAM0KJ Google Ads cpc 228.80 6 7 0.2143 49.03
11 order_K9WAM0KJ Google Ads cpc 228.80 7 7 0.2500 57.20
12 order_OEVGKH25 Meta paid_social 557.36 1 9 0.0222 12.39
13 order_OEVGKH25 Meta paid_social 557.36 2 9 0.0444 24.77
14 order_OEVGKH25 Meta paid_social 557.36 3 9 0.0667 37.16
15 order_OEVGKH25 Google Ads cpc 557.36 4 9 0.0889 49.54
16 order_OEVGKH25 Google Ads cpc 557.36 5 9 0.1111 61.93
17 order_OEVGKH25 Google Ads cpc 557.36 6 9 0.1333 74.31
18 order_OEVGKH25 Google Ads cpc 557.36 7 9 0.1556 86.70
19 order_OEVGKH25 Google Ads cpc 557.36 8 9 0.1778 99.09
20 order_OEVGKH25 Google Ads cpc 557.36 9 9 0.2000 111.47
21 order_ZJ396YFI Meta paid_social 603.77 1 3 0.1667 100.63
22 order_ZJ396YFI Meta paid_social 603.77 2 3 0.3333 201.26
23 order_ZJ396YFI Meta paid_social 603.77 3 3 0.5000 301.89
24 order_QWQMKI0 Meta paid_social 562.06 1 9 0.0222 12.49

=== VALIDATION CHECK ===
total_time_decay_revenue
0 28353.55
```

What You're Seeing

Channels appearing later in journeys gained significantly more attribution value.

Examples:

- Google Ads strengthened further,
- Direct traffic increased,
- Meta weakened relative to first-touch.

Why This Matters

Time-decay attribution assumes:

- recency influences behavioural impact.

This often increases visibility into:

- branded search,
- conversion-focused campaigns,
- and lower-funnel activity.

Important Limitation

Recency does not guarantee influence.

Later touchpoints may simply:

- capture intent already created elsewhere.

Strategic Interpretation

Time-decay attribution tends to strengthen channels appearing closer to conversion behaviour.

In this dataset:

- Google Ads,
- Direct traffic,
- and other lower-funnel interactions

gained more attribution value as touchpoints moved closer to purchase.

This reflects a common real-world pattern:
users often:

- discover a brand earlier,
- then later return through search,
- direct visits,
- or branded navigation when ready to convert.

For marketers, this model can be useful when:

- recency appears behaviourally meaningful,
- buying intent strengthens over time,
- or lower-funnel optimisation is a priority.

However, this also creates risk.

Channels responsible for:

- awareness,
- discovery,
- or repeated exposure

may appear artificially weak if their influence occurs earlier in the customer journey.

Step 9 — Compare Attribution Models

At this stage, we now have several attribution models built from the exact same customer journeys.

Now comes the most important part:
comparing them side-by-side.

This is where we begin to see how different attribution assumptions completely change the marketing story.

The conversions themselves never changed.

Only the way we distributed credit changed.

This final step helps reveal:

- which channels are highly sensitive to attribution models,
- which channels remain stable,
- and how different reporting approaches can lead marketers toward very different strategic decisions.

Script

Python

`09_compare_attribution_models.py`

Goal

Compare all deterministic models side-by-side.

What This Script Is Doing

This script consolidates:

- first-touch,
- last-touch,
- linear,
- position-based,
- and time-decay attribution

into a unified comparison framework.

It measures:

- attribution sensitivity,
- model spread,
- and changing channel valuation.

Output

```
C:\Users\kyler\k2\Projects\local-attribution-sandbox>python scripts/09_compare_attribution_models.py

=== ATTRIBUTION MODEL COMPARISON ===
source      medium  first_touch_revenue  last_touch_revenue  linear_revenue  position_based_revenue  time_decay_revenue  max_model_revenue  min_model_revenue  model_spread  Discovery  likely_channel_role
0  Meta  paid_social  13664.88  6587.40  8963.64  9823.89  7643.10  13664.88  6587.40  7077.48  Conversion / demand generation
1  Google Ads  cpc  7197.66  10917.85  9843.84  9226.34  10521.37  10917.85  7197.66  3810.19  Conversion capture / late-stage intent
2  Direct  (none)  6296.44  7456.31  7192.49  6934.59  7321.16  7456.31  6296.44  1159.87  Conversion capture / late-stage intent
3  Organic  organic  376.96  2080.14  1227.76  1201.63  1552.45  2080.14  376.96  1623.18  Conversion capture / late-stage intent
4  Unknown  (not set)  987.61  1391.85  1215.72  1167.11  1315.49  1391.85  987.61  404.24  Conversion capture / late-stage intent

=== MODEL TOTAL VALIDATION ===
first_touch_total  last_touch_total  linear_total  position_total  time_decay_total
0  28353.55  28353.55  28353.45  28353.86  28353.55
```

What You're Seeing

The same conversions generated completely different strategic narratives depending on attribution assumptions.

Examples:

Meta / paid_social

Spread:

~\$7k

Meta appears:

- extremely attribution-sensitive,
- and highly dependent on attribution philosophy.

Google Ads / cpc

Very stable across:

- last-touch,
- time-decay,
- and linear attribution.

This suggests:

- strong lower-funnel,
- conversion-capture behaviour.

Direct / (none)

Direct strengthened heavily under:

- last-touch,
- and time-decay.

But this remains strategically ambiguous.

Direct traffic may represent:

- genuine brand recall,
- or attribution leakage.

Core Attribution Insight

***The conversions never changed.
Only the attribution assumptions changed.***

This is the foundational lesson of attribution modelling.

Different models:

- reshape perceived channel value,
- alter optimisation priorities,
- and create different strategic narratives from identical underlying data.

Important Limitation

No attribution model in this workflow proves:

- why a user converted,
or:
- which touchpoint truly caused behavioural change.

The models only redistribute observable conversion value using different attribution rules.

This means the same customer journey can produce:

- different optimisation decisions,
- different channel valuations,
- and different strategic conclusions

without the underlying conversions ever changing.

Understanding that distinction is one of the most important skills in marketing measurement.

Strategic Interpretation

The goal of attribution modelling is not:

- finding the “correct” model.

The real goal is understanding:

- behavioural ambiguity,
- measurement limitations,
- and how analytical assumptions shape business decisions.

Key Observations From The Attribution Comparison

1. Meta is highly attribution-sensitive

None

Spread: ~7k

This means:

- Meta's perceived value changes significantly depending on attribution philosophy,
- especially between first-touch and last-touch approaches.

This is a common characteristic of:

- awareness,
- discovery,
- and demand-generation channels.

Under first-touch attribution, Meta appears significantly stronger because it often introduces users earlier in the customer journey.

Under last-touch attribution, its perceived value weakens because another channel often captures the final conversion interaction.

This demonstrates an important real-world tension:

- marketing teams often defend upper-funnel channels using multi-touch logic,
- while finance or reporting teams may rely heavily on last-click reporting.

The same underlying conversions can therefore produce very different strategic narratives.

Business Risk

If an organisation relies heavily on last-click attribution, upper-funnel demand generation may appear unprofitable and receive budget cuts despite contributing significantly earlier in journeys.

2. Google Ads behaves like a demand-capture channel

Google Ads remained relatively stable across:

- last-touch,
- time-decay,
- and linear attribution.

This pattern suggests Google Ads is functioning primarily as:

- conversion capture,
- intent resolution,
- and lower-funnel demand harvesting.

This is common in search behaviour.

Users often:

- discover brands elsewhere,
- then later use search to validate, compare, or complete conversion behaviour.

This is an important distinction:

Search may not always create demand.

Often, it captures demand already created elsewhere.

Business Risk

Channels specialising in demand capture may appear disproportionately valuable if awareness-generating activity is not measured properly.

3. Direct traffic remains strategically ambiguous

Direct traffic strengthened significantly under:

- last-touch,
- and time-decay attribution.

At first glance, this may suggest:

- strong conversion behaviour,
- brand familiarity,
- or returning user intent.

However, Direct traffic should always be interpreted cautiously.

Direct can represent:

- genuine brand recall,
- returning customers,
- dark social,
- unattributed traffic,
- tracking leakage,
- or broken campaign tagging.

This makes Direct one of the most dangerous channels to over-interpret in attribution analysis.

Business Risk

Over-crediting Direct traffic can create false confidence in brand strength while masking underlying attribution leakage or poor campaign tagging.

4. Organic behaving late-stage is interesting

Organic traffic strengthened more heavily under:

- last-touch,
- and time-decay models.

This may suggest users are:

- revisiting organically,
- validating decisions,
- researching before purchase,
- or using branded search behaviour later in the funnel.

This highlights how attribution models can reveal behavioural patterns beyond simple channel performance reporting.

The Most Important Attribution Insight

The underlying conversions never changed.

Only the attribution assumptions changed.

This is one of the most important lessons in attribution modelling.

Different attribution models:

- distribute credit differently,
- prioritise different behavioural assumptions,
- and therefore produce different strategic narratives from the exact same customer journeys.

That does not make attribution useless.

But it does mean attribution should be treated as:

- an interpretive framework,
not:
- objective truth.

The goal is not finding a “perfect” attribution model.

The goal is understanding:

- what assumptions each model encodes,
- what behaviours each model emphasises,
- and how those assumptions influence strategic decision-making.

Why Strong Measurement Workflows Matter

Throughout this workflow, the system repeatedly validated:

- backend revenue,
- duplicated touchpoints,
- identity stitching quality,
- conversion reconciliation,
- and attribution outputs.

This matters because attribution problems are often not modelling problems first.

They are:

- data quality problems,
- identity problems,
- stitching problems,
- and reconciliation problems.

Without validation, attribution systems can silently:

- duplicate revenue,
- inflate performance,
- misattribute conversions,
- and distort optimisation decisions.

Strong marketing measurement depends just as much on:

- skepticism,
- validation,
- and reconciliation

as it does on attribution modelling itself.

Final Reflection

This project started as a way to better understand why marketing reporting often felt directionally correct, but still somehow incomplete.

Working through these exercises helped me better understand how attribution models shape the way marketers interpret performance data.

Hopefully it helped you think more critically about those systems as well. Good luck.